

A Reference-Free Evaluation Funnel for Japanese News Summaries, Delivered Twice as an Installable Plugin

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50 XL-Sum Japanese articles, 250 candidate summaries

Abstract—A summariser in production is handed an article and returns a summary; nobody hands it a human reference. This report describes an evaluator built to that contract. Exploration showed the 250 supplied candidates are not a quality gradient but five distinct populations, and that the two most dangerous—fluent text that reverses the article, and fluent text that invents its central claim—are invisible to every cheap signal, because they are on topic by construction. The design separates constraints a script can prove from constraints only a reader can judge, and requires an independent reviewer to confirm every stopping decision. Validation rests on seven checks that do not share a failure mode, including a natural experiment in which the same candidate texts appear under both their own and a foreign article, and a third independent read of the 109 candidates for which no ground truth can be derived at all. The evaluator ships as an installable plugin, implemented twice on two agent hosts; the two implementations agree on routing for 94.4% of candidates and on terminal category for 74 of 74.

Index Terms—LLM evaluation, reference-free metrics, summarisation quality, agent isolation, auditability, reproducibility

I. EXPLORATION

A. First contact: what the 250 candidates actually are

The dataset is 50 articles (483–3664 characters, median 1587) with a human reference each (41–164 characters, median 93), and five opaque candidates per article (22–302 characters, median 125).

Reading candidates side by side inside their articles, rather than as a flat list, made the structure obvious within the first few articles: the five candidates per article are not five samples from one generator. They are draws from distinct populations. A string-level scan of the corpus alone—no model, no scoring—confirmed and counted them (Table I). Four populations are mechanically derivable; the fifth is the remainder and is the interesting one.

Three structural facts came out of the same scan and shaped everything downstream.

The sentence limit is almost never the problem. Only three of 250 candidates exceed three sentences, and all three are article copies that would be stopped anyway. A rule that enforces the stated product specification is necessary but will almost never fire; treating “ ≤ 3 sentences” as the headline quality dimension would have been a serious misallocation of effort.

The corpus contains an accidental controlled experiment. Twenty groups of candidates share byte-identical text after

TABLE I
CORPUS CENSUS. CATEGORIES DERIVED FROM THE CORPUS BY STRING COMPARISON ONLY, WITH NO EVALUATOR OUTPUT INVOLVED. USED AT THE VERY END AS WEAK LABELS, NEVER DURING SCORING.

Population	<i>n</i>	Derivation
Reference reproduced	58	candidate = reference
Copy of the article	50	candidate $\subset$ article body
Reference fragment	17	reference starts with candidate
Belongs elsewhere	16	matches a <i>different</i> article
Generated, unlabelled	109	none of the above
Total	250	

Unicode normalisation. Eight of those groups sit inside a single article—the same text scored twice, independently. Fourteen span articles: the same text is attached to its own article in one place and to a foreign article in another. This is a free reliability instrument and Section III uses it as one.

The references are not ground truth, and the data says so. The assignment warns that XL-Sum reference quality is uneven. It is worse than uneven: several references assert facts the article body never contains. The reference for article 35278497 states a 4 割 sexual-assault share, a suspect profile, and a 379-case figure, none of which appear in the supplied text; the reference for features-and-analysis-46643494 is a photo caption that opens with この写真, a demonstrative whose antecedent exists only in an image nobody is given. Any design that scores a candidate by its distance to the reference would reward reproducing these.

B. Hypotheses, and which ones died

Five hypotheses were formed before building anything. Two survived unchanged, one survived in weakened form, and two were killed by the data (Table II).

The last row is the finding the rest of the design turns on. Inspecting the 109 unlabelled candidates one by one exposed a failure mode with no detector anywhere in the pipeline as it then stood: text that is fluent, within the sentence limit, not a copy, not truncated, on topic, and *states the opposite of the article*. Article 41396293 reports the Japanese prime minister announcing a dissolution of the lower house; its candidate 0c954428 reports him announcing that he will *not* dissolve it. Article 34991666 reports two attackers dead; candidate

TABLE II
HYPOTHESES FORMED DURING EXPLORATION AND THEIR FATE.

Hypothesis	Outcome
Copying is detectable with string evidence, not embeddings	Survived. A copy is maximally similar to its article, so similarity is exactly the wrong instrument. Containment and n -gram coverage caught 50 of 50.
Embedding similarity separates topic mismatch	Survived. Two disjoint 10-article probes: matched 0.707–0.868, mismatched 0.163–0.398.
Similarity can screen relevance on its own	Weakened. On all 250 pairs the bands overlap: off-topic reaches 0.4813, on-topic falls to 0.4034. It may nominate; it may not decide.
Missing sentence-final punctuation indicates truncation	Killed. 体言止め—a verbal noun closing a headline—is standard Japanese news register and looks identical to an amputated predicate.
Factual defects form one gradient, so one faithfulness score suffices	Killed. A wrong peripheral number and a reversed main event are not two points on a scale; one is a flawed summary, the other is a false one.

f6f8e339 reports them fled to another state and still at large. Nothing upstream can see this. The string evidence is clean. The embedding similarity is *high*, because the topic is right—that is precisely why the failure is dangerous. A sibling mode invents rather than inverts: candidate 41875333_aa36a21a attributes to the US president the quoted line 「米国はもはや世界の警察ではない」, which appears nowhere in the article and displaces the claim the headline actually makes.

Set against these, a case such as 40490051_5d9feef is a different animal: it substitutes five details (シリア for イラク, 男性 for 女性, 30 代 for 10 代, 8000 for 6000, 500 for 300) while the central event—a fierce IS-clearing battle in Mosul’s old city—is correctly reported. It is a bad summary, not a false one, and an evaluation that collapses both into “score zero” has thrown away the ranking the application needs.

II. DESIGN

A. The evidence boundary

The runtime evaluator sees exactly one article and one candidate—never a reference, never another article, never a corpus. This is not a stylistic preference: a signal that needs a second article is unavailable to a caller who sent one request, and a signal derived from a reference is unavailable to an application whose purpose is to run where no reference exists. An earlier relevance gate ranked each candidate against every article in the batch; that evidence was removed even though it worked, because it measured something the deployed system cannot. References enter once, after every score is final, in Section III.

B. Two layers of hard constraint, then a graded score

Figure 1 is the funnel, annotated with the actual counts from the submitted run. The organising principle is that constraints divide by what kind of evidence can settle them.

Layer 1 is physical. Empty output, sentence count above three, and near-verbatim copying are settled by a script,

TABLE III
SOFT RUBRIC, APPLIED ONLY TO CANDIDATES THAT SURVIVE BOTH HARD LAYERS. THE FOUR DIMENSIONS SUM TO THE TOTAL EXACTLY.

Dimension	Max	Question	Mean
Faithfulness	50	Is every claim supported by the article?	38.7
Coverage	30	Is the main event and its key facts present?	19.0
Coherence	15	Complete, grammatical, ordered?	13.5
Conciseness	5	Compact, without repetition?	4.8
Total	100	76.1 over 165 survivors	

because the measurement *is* the conclusion: a containment ratio near 1.0 is not evidence of copying, it is copying.

Layer 2 is semantic. Whether a candidate is grounded in its article cannot be settled by any measurement, so a dedicated Grounding Gate agent screens every survivor before any scoring happens. It answers one question—is this candidate grounded?—and returns at most one central category (OFF_TOPIC, FACTUAL_REVERSAL, FABRICATED_CONTENT) with one article span and one candidate span as evidence. It is deliberately blind to quality: it never receives the rubric, never sees the anchor, and cannot emit a score. An agent that both screens and scores will trade a central defect for a low score instead of stopping it.

Between the two layers sits embedding relevance, which is *required but never terminal*. Every survivor must leave that stage carrying a similarity value against its own article; a run that could not reach the embedding model is incomplete, not cheaper. But a value below threshold is only a hint that the semantic layer should look closely.

Survivors of both layers receive a graded score (Table III). Weights follow directly from the exploration: faithfulness carries half the total because fluent-but-false is the dominant risk in this corpus, and conciseness carries five points because the sentence limit is already a hard constraint and almost never violated.

Terminal outcomes all score 0 but keep separate rank tiers, so five candidates remain orderable even when several fail. Reversal, fabrication and off-topic share the worst tier, because each misleads a reader who cannot check the source; copying and truncation rank above them, because a copy at least contains true sentences.

C. Independent review of every stopping decision

No stage in the funnel may stop a candidate on its own authority. Each proposal—from the script, from the grounding gate, from the scorer’s backstop—goes to a Reviewer in a fresh context that re-derives the evidence from the article. The review is not a rubber stamp: in the submitted run it rejected 2 of 53 physical proposals, 8 of 37 grounding proposals (including all eight fabrication claims), and 3 of 4 late scorer proposals, sending each rejected case onward to be scored rather than stopped. It also escalated four truncations the gate had not proposed. Disagreement that survives one revision round is preserved as low confidence rather than averaged away.

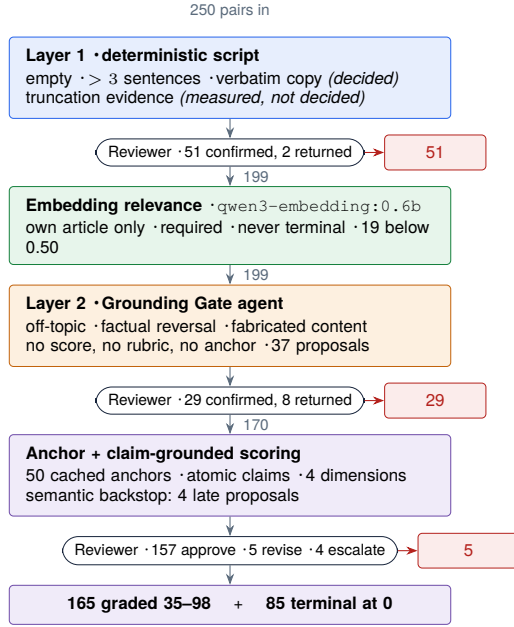


Fig. 1. The funnel, annotated with counts from the submitted 250-pair run. Red exits are Reviewer-confirmed terminations. The 51 pairs stopped at layer 1 cost one script pass and one confirmation; 80 of 250 never reach anchor generation and scoring, the most expensive stage.

D. Alternatives considered and rejected

Table IV lists the approaches that were tried and dropped, each with the measurement that killed it. Two deserve a sentence. A cross-encoder reranker (bge-reranker-v2-m3) separates the off-topic population more cleanly than the embedding does in mean terms—0.0001 against 0.9300—but at a 0.50 cut it flags 15 legitimate candidates as unrelated where the embedding flags 3, so it buys no recall and costs five times the false alarms. And embedding the candidate against the generated anchor correlated better with coverage ($\rho = 0.630$ versus 0.528) yet predicted it worse under leave-one-out (MAE 4.01 versus 3.62); a feature that correlates better and predicts worse is a feature fitted to the sample, so the anchor stayed a reasoning aid and never became a number.

E. Delivery: the design is a plugin, not a script

The evaluator is packaged as an installable plugin so that anyone who needs it can run it rather than reconstruct it. The primary implementation is a Claude Code plugin: one slash command, one skill, four isolated agents (Grounding Gate, Scorer, Reviewer, Reporter) and the deterministic scripts they call. Installation is two commands against a local marketplace declared in the repository. There is no orchestration notebook, no prompt to paste and no hidden state—the manifest, rubric, few-shot bank and agent contracts are all files under version control, so a reviewer can read exactly what each role was told.

The same design is implemented a second time as a Codex plugin. This is not redundancy for its own sake—it is the instrument for the cross-host check in Section III-

TABLE IV
REJECTED APPROACHES AND THE EVIDENCE THAT REJECTED THEM.

Approach	Why it was rejected
Score against the reference	Unavailable in production, and demonstrably wrong here: 45 of 50 references were outscored by a generated candidate in their own article.
Similarity as a quality score	A verbatim copy is the most similar text there is.
Cross-encoder reranker	Same recall (16/16), five times the false positives (15 vs. 3 of 234) at a 0.50 cut.
Anchor similarity as a scoring feature	Higher correlation, worse leave-one-out error.
Batch-relative relevance	Uses articles a production caller never sent. Removing it changed 0 of 19 nominations.
Regex decides truncation	体言止め is indistinguishable from an amputated predicate without the deleted text.
One agent screens and scores	It trades a central defect for a low score instead of stopping.

TABLE V
SEVEN VALIDATION CHECKS ON THE SUBMITTED RUN. “A” IS THE PRIMARY CLAUDE-PLUGIN RUN, “B” THE INDEPENDENT CODEX-PLUGIN RUN.

Check	What it tests	Result
Arithmetic & bands	500 records, sums and label boundaries	0 errors
Repeated text	8 duplicate pairs inside one article	8/8, 8/8
Pair conditioning	14 texts under own <i>and</i> foreign article	14/14, 14/14
Weak labels	141 corpus-derivable categories	.915 / .901
Within-article order	planted failure vs. its reference	0/50, 0/50
Cross-host routing	A vs. B on 250 pairs	.944
Third read	109 candidates with no ground truth	ρ .960

F. Both plugins emit the same record schema, including an `evaluation_trace` that lists every stage visited with its evidence, so any score in `scores.jsonl` can be replayed backwards to the measurement that produced it.

III. VALIDATION

An evaluation is only worth its weakest justification, so the goal here was several checks that fail differently rather than one check repeated. Table V is the summary; the subsections give the reasoning.

A. Internal consistency

Across the 500 records of both runs, every soft score equals the sum of its four dimensions and every quality label falls inside its declared band; there are zero arithmetic and zero banding violations. This is a floor, not a result—it says the machinery did what it claims, nothing about whether the judgements are right.

B. The same text, scored twice

Eight groups of candidates are byte-identical *and* attached to the same article. Each was scored independently, in a separate context, without either scorer knowing the twin existed. All eight received identical totals *and* identical values on all four dimensions, in both implementations. Run-to-run variance on repeated input is therefore not the reason for any spread reported below.

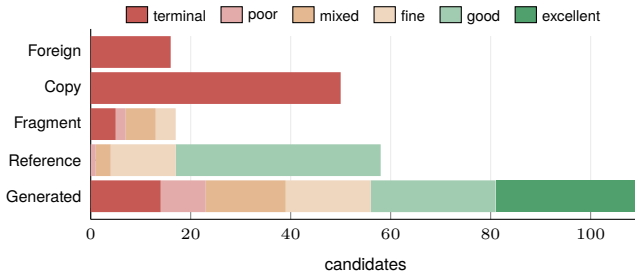


Fig. 2. Outcome of the primary run by corpus-derived population. The two populations that should never survive never do; the fragment population is graded rather than stopped; and no reference reproduction reaches the top band.

C. A natural experiment in pair conditioning

Fourteen further groups are byte-identical but split across articles: the same text sits under its own article in one place and under a foreign one in another. If the evaluator were grading text quality, it would give both copies the same score. It does not. In all 14 groups, in both implementations, the foreign placement was terminated at 0 as off-topic and the native placement was graded on its merits—for instance the text of 54083932_9e7163bb scores 87 under its own article and 0 as 44708203_6d931562 under another. The evaluator is conditioning on the pair, which is what the contract requires and what a text-only metric cannot do.

D. Agreement with corpus-derived weak labels

After both runs were final, references were opened for the first time and the Table I categories were used as weak labels. Figure 2 shows the outcome distribution per population. All 50 article copies and all 16 foreign-article candidates terminated, in both runs. No reference reproduction was ever terminated. Overall agreement on the 141 labellable candidates is 129/141 (0.915) for run A and 127/141 (0.901) for run B.

Two things in that figure matter more than the headline number. First, the fragment population is graded rather than uniformly stopped: 5 of 17 terminate and the remaining 12 land between 39 and 70, none reaching GOOD. That is a deliberate divergence from the label, discussed in Section IV, not a detection failure. Second, *no* reference reproduction reached EXCELLENT while 28 generated candidates did. This is the reference-quality problem made quantitative: the evaluator credits only what the article body supports, and XL-Sum references routinely assert more than that.

E. Within-article ordering

The application needs candidates ordered inside an article, so ordering was checked directly rather than inferred from scores. Across all 50 articles, in both runs, *zero* articles placed a planted failure at or above the reference reproduction for that article. Ordering is also not degenerate: the median spread between the best and worst surviving candidate within an article is 26 points (maximum 61), and every article retained between two and four survivors, so no article was either wiped out or passed untouched.

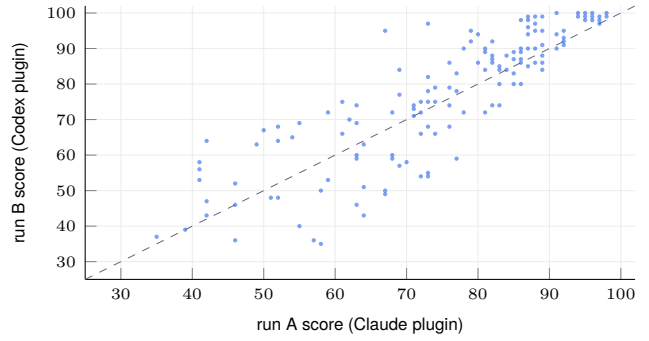


Fig. 3. The 162 candidates graded by both implementations. $\rho = 0.891$, mean absolute difference 6.85 points. The dashed line is equality.

F. Two implementations, two hosts

The strongest available check on a design—as opposed to a run—is to implement it twice and compare. The Codex plugin was written against the same rubric but a different host, different agent runtime and different underlying model, and neither implementation saw the other’s output.

On 250 pairs the two agree on routing (terminal versus graded) for 236, or 94.4%, and on *which* terminal category, for all 74 pairs both stopped. On the 162 pairs both graded, Spearman $\rho = 0.891$ with a mean absolute difference of 6.85 points and a signed mean of -1.97 (Fig. 3).

The disagreements are more informative than the agreements, and they are one-sided in a way that is easy to explain. All 11 pairs stopped only by A are severity calls on a defect B also found: 6 reversals, 4 fragments and 1 fabrication, which B scored between 28 and 66—low, but not stopped. The 3 stopped only by B were scored 58–71 by A. Not one disagreement is a case where one implementation saw a defect the other missed entirely. The two systems disagree about where the cliff is, not about which candidates are near it.

Exact agreement on the five-band label is only 88 of 162 (54.3%), and that number is reported here because it is the least flattering result in the report. It is a banding artefact—a mean absolute difference of 6.85 straddles boundaries that are 10 to 15 points wide—but it is a real caution: the bands are convenient for a reader and should not be treated as stable, whereas the underlying ordering is.

G. The block with no ground truth

Every check so far rests on the 141 candidates the corpus can label. But those are the easy ones: copies, fragments and misfiled text are detectable by construction. The 109 generated candidates—the block that actually carries the fluent factual errors—have no derivable label at all, and they are 44% of the corpus. An evaluation that reports 0.915 agreement while remaining silent about this block is reporting the wrong number.

So the 109 were scored a third time, by a separate reading that used the same rubric but had no access to either run’s output. Against the primary run this third read gives $\rho = 0.960$ with a mean absolute difference of 4.09 points; against the

secondary run, $\rho = 0.936$ and 5.95 points. More pointedly: the 14 candidates the primary run terminated in this block were given a mean of 32.5 by the independent read (range 20–53) against 77.0 for the rest. An independent reader who was never told those candidates had been stopped put every one of them at the bottom of the distribution.

This does not make the scores correct—three readings of the same rubric can share a bias, and none of them is a human judgement. What it does establish is that the signal on the unlabelled block is reproducible rather than idiosyncratic, which is the claim that was actually missing.

H. Reproducing the numbers in this report

Every quantity quoted above is re-derived from the shipped artifacts by `code/validation/verify_report_numbers.py`, which reads only `data/`, the two run files and the third-read file, and asserts each claim. It prints 91 checks and exits non-zero if any of them moves. A reader who does not trust a number in this report can recompute it in one command rather than take it on faith.

IV. LIMITATIONS

A. What the evaluation does not measure

No human judgement enters anywhere. Every agreement number in Section III is either agreement with a mechanically derived label or agreement between readings of the same rubric, and neither is accuracy. The honest ceiling on all of it is *reproducible*, not *correct*.

Reference reproductions are treated as acceptable by the weak-label check, which Section I showed is not safe: the worst of them scores 42 and the evaluator was right to say so. The label and the evaluator disagree there, and the label is wrong.

It also measures nothing a reader would call style, register or audience fit: beyond grammatical completeness it has no view on whether a summary is well written, nor on the newsworthiness of the facts chosen—only on whether they are the article’s main ones.

B. What it measures poorly

Truncation is the weak point, and the failure is partly structural. A one-record probe made this concrete: candidate `51395937_e409196f` is the reference cut from 「…方針を発表した。」 to 「…方針を発表」. The Reviewer declined to call it truncated, arguing that 発表 is a verbal noun discharging the clause and that this is ordinary headline register. The argument is sound and the conclusion is wrong—the corpus shows the deleted characters were した. Nothing in the article or the candidate distinguishes 体言止め from an amputated predicate; only the deleted text does, and a reference-free evaluator does not have it. The fragment population therefore splits in two: endings amputated mid-token (しかし、女性 2 人が車、救助し) are visible and missing them is a recall problem worth fixing, while 体言止め endings are undecidable under the contract and should not be counted

against the evaluator. Reporting both as one category, as the 5-of-17 figure does, overstates the deficiency.

The five-band labels are less stable than the scores, as Section III-F showed. Bands are for readers; ordering is for decisions.

Cost is uneven. The funnel settles 51 of 250 pairs with a script and one confirmation pass and keeps 80 of 250 out of anchor generation and scoring entirely, but the price of moving the truncation judgement from the script to the Reviewer is that a fragment now consumes both before it is stopped—about six percent of the scoring work in this run.

C. One structural gap, stated plainly

The Reviewer in the submitted run held file-read tools and the corpus sat on disk with its reference column intact. Its inputs contained no reference field and the artifacts show none was used, but that is a guarantee by convention, and a convention is not evidence. A structurally blind Reviewer that receives article and candidate inline and holds no read tools is written and shipped in both plugins; no run has used it yet. The right next step is to re-adjudicate the 85 terminal decisions with it and publish the delta, because that number is the honest measure of how much the reachability of a reference mattered.

D. What I would do next

More data, first and above everything else. The whole report rests on 250 pairs from 50 articles of one publisher in one language, with a median article length of 1587 characters. Every threshold here—the 0.50 similarity cut, the four dimension maxima, the five band boundaries—is calibrated to that sample and should be assumed invalid outside it. The immediate work is to run the same funnel over a much larger Japanese corpus drawn from several publishers, then across languages, because a design whose hard constraints include Japanese sentence segmentation and 体言止め register has obvious language-specific parts that must be re-derived rather than translated. Only after that does it make sense to spend on human adjudication of a stratified sample, controlled perturbations for numbers, entities, negation and causality, and re-calibration of the embedding threshold per model and domain.

V. CONCLUSION

This is a working, fully audited prototype with evidence on one corpus: 250 pairs routed, 85 stopped by a confirmed terminal decision, 165 graded between 35 and 98, no arithmetic error, no ordering inversion against a reference, 94.4 % routing agreement with an independently implemented twin, and a reproducible signal on the 44 % of the corpus where no ground truth exists. It is not a production accuracy guarantee, for the reason in Section IV-D: one corpus, one language, one publisher. What the design provides is the property that makes the next corpus cheap—it installs as a plugin, and every score carries the trace that produced it.